Bayesian Modelling:
Bayesian Inference for Gaussian Models

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[0] C. P. Robert. The Bayesian Choice. Springer, 2007.

Outline

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Gaussian model with known variance (recap)

Gaussian model with known variance

- Consider $n = 9$ measurements of wing lengths of midges:

$$\mathbf{x} = (1.64, 1.70, 1.72, 1.74, 1.82, 1.82, 1.90, 2.08).$$

- We model these as independent draws:

$$x_i = \mu + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2) = N(0, 0.13^2).$$

- Likelihood:

$$p(\mathbf{x}|\mu) \propto \exp\left(-\frac{1}{2} \sum_{i=1}^{n} \frac{(x_i - \mu)^2}{\sigma^2} \right).$$

Gaussian model with known variance

- We set a conjugate prior:

$$\mu \sim N(m_0, \sigma_0^2).$$

- Posterior:

$$p(\mu|\mathbf{x}) \propto \exp\left(-\frac{1}{2} \sum_{i=1}^{n} \frac{(x_i - \mu)^2}{\sigma^2} - \frac{(\mu - m_0)^2}{2\sigma_0^2} \right)$$

$$\propto \exp\left(-\frac{1}{2} \frac{(\mu - m_n)^2}{\sigma_n^2} \right),$$

$$m_n = \sigma_n^2 \left(\frac{m_0}{\sigma_0^2} + \frac{n}{\sigma^2} \bar{\mathbf{x}} \right),$$

$$\sigma_n^2 = \frac{1}{\sigma_0^2} + \frac{n}{\sigma^2}.$$

Gaussian model with unknown variance

Gaussian model with unknown variance

- Model the midge wing measurements as independent draws:

$$x_i = \mu + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2).$$

- Set conjugate priors:

$$\mu \sim N(m_0, \sigma_0^2),$$
$$\sigma^2 \sim \Gamma^{-1}(\alpha, \beta),$$

  where Γ^{-1} is the inverse gamma distribution.

- Likelihood:

$$p(\mathbf{x}|\mu, \sigma^2) \propto \exp\left(-\frac{1}{2} \sum_{i=1}^{n} \frac{(x_i - \mu)^2}{\sigma^2} \right).$$

Inverse gamma distribution

- $X \sim \Gamma^{-1}(\alpha, \beta)$ has respective density, mean, and variance

$$p(x|\alpha, \beta) = \frac{\beta^\alpha}{\Gamma(\alpha)} \frac{1}{x^{\alpha+1}} \exp(-\beta/x),$$

$$\mathbb{E}[X] = \frac{\beta}{\alpha - 1} \quad \text{if } \alpha > 1,$$

$$\text{Var}(X) = \frac{\beta^2}{(\alpha-1)^2(\alpha-2)} \quad \text{if } \alpha > 2.$$

- If $X \sim \Gamma(\alpha, \beta)$ then $1/X \sim \Gamma^{-1}(\alpha, \beta)$.

Reparametrisation

- Expert opinion suggests that μ and σ^2 are not far from 1.9 mm and 0.1^2 mm.
- Reparametrise for ease of interpretation:

$$\mu|\sigma^2 \sim N(m_0, M_0\sigma^2),$$
$$\sigma^2 \sim \Gamma^{-1}(\nu_0/2, \sigma_0^2\nu_0/2).$$

- We set $m_0 = 1.9$, $M_0 = 1$, $\nu = 2$ and $\sigma_0^2 = 0.01$.

Gaussian model with unknown variance

- On the exercise sheet you'll show that

$$\sigma^2|\mathbf{x}_{1:n} \sim \Gamma^{-1}(\nu_n/2, \sigma_n^2\nu_n/2),$$
$$\mu|\sigma^2, \mathbf{x}_{1:n} \sim N(m_n, M_n\sigma^2),$$

where

$$M_n = \frac{M_0}{M_0 n + 1},$$
$$m_n = M_n\left(\frac{m_0}{M_0} + n\bar{\mathbf{x}}\right),$$
$$\nu_n = \nu_0 + n,$$
$$\sigma_n^2 = \frac{1}{\nu_n}\left(\nu_0\sigma_0^2 + \sum_{i=1}^{n}(x_i - \bar{\mathbf{x}})^2 + \frac{(\bar{\mathbf{x}} - m_0)^2}{M_0 + 1/n}\right).$$

Posterior means

- The conditional posterior $\mu|\sigma^2, \mathbf{x}_{1:n}$ is Gaussian, but it has mean m_n for every value of σ^2.
- Hence the marginal posterior mean of μ is also

$$\mathbb{E}[\mu|\mathbf{x}] = M_n\left(\frac{m_0}{M_0} + n\bar{\mathbf{x}}\right) = \frac{1/M_0}{1/M_0 + n} m_0 + \frac{n}{1/M_0 + n}\bar{\mathbf{x}}.$$

- It is (again) a weighted average of the prior mean and sample mean.
- $1/M_0$ can be interpreted as a number of virtual samples representing the degree of confidence in the prior.
- The posterior mean of σ^2 is

$$\mathbb{E}[\sigma^2|\mathbf{x}_{1:n}] = \frac{\sigma_n^2 \nu_n}{\nu_n - 2}.$$

Linear regression (recap)

Regression

- Goal: explain dependent variable (or response) y in terms of independent variables (or covariates) $x_1, \ldots, x_k$.
- Linear regression:

$$y_i = \beta_0 + \beta_1 x_{i1} + \ldots + \beta_k x_{ik} + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2).$$

- We will use the more compact

$$\mathbf{y} = \mathbf{X}\beta + \varepsilon, \quad \varepsilon \sim N(0, \sigma^2 I_{n \times n}).$$

Least squares

- We can estimate β by minimising the residual sum of squares:

$$\text{RSS}(\beta) = (\mathbf{y} - \mathbf{X}\beta)^T(\mathbf{y} - \mathbf{X}\beta).$$

- When $\mathbf{X}$ has full rank so that $\mathbf{X}^T\mathbf{X}$ is invertible,

$$\widehat{\beta}_{\text{LS}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}.$$

- $\widehat{\beta}_{\text{LS}}$ is unbiased:

$$\mathbb{E}[\widehat{\beta}_{\text{LS}}] = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbb{E}[\mathbf{y}] = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}\beta = \beta.$$

- Standard errors can be obtained using

$$\widehat{\text{Cov}}(\widehat{\beta}_{\text{LS}}) = \widehat{\sigma}^2(\mathbf{X}^T\mathbf{X})^{-1},$$

where $\widehat{\sigma}^2$ is an estimate of σ^2.

Gaussian errors

- The least squares estimator does not rely on distributional assumptions.
- Assuming independent, Gaussian errors yields the likelihood:

$$L(\beta, \sigma^2 | \mathbf{y}) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\Big(-\frac{1}{2\sigma^2}(\mathbf{y} - \mathbf{X}\beta)^T(\mathbf{y} - \mathbf{X}\beta)\Big).$$

- Then $\widehat{\beta}_{\mathsf{LS}}$ is the MLE for β.
- The MLE for σ^2 is biased. An unbiased estimator is

$$\widehat{\sigma}^2 = \frac{1}{n - (k+1)}(\mathbf{y} - \mathbf{X}\widehat{\beta}_{\mathsf{LS}})^T(\mathbf{y} - \mathbf{X}\widehat{\beta}_{\mathsf{LS}}).$$

Prediction

- Recall $\mathbb{E}[\mathbf{y}] = \mathbf{X}\beta$.
- Substituting in $\widehat{\beta}_{\mathsf{LS}}$ yields

$$\widehat{\mathbf{y}} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y} =: \mathbf{H}\mathbf{y},$$

  where $\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ is called the hat matrix since it puts the hat on $\mathbf{y}$.
- Predicted residuals can be obtained in terms of the hat matrix as well:

$$\widehat{\varepsilon} = \mathbf{y} - \widehat{\mathbf{y}} = \mathbf{y} - \mathbf{X}\widehat{\beta}_{\mathsf{LS}} = (I - \mathbf{H})\mathbf{y}.$$

- Predicted residuals are a diagnostic: no trend and resemblance to $N(0, \sigma_e^2)$-distributed samples are evidence of good model fit.
- σ_e^2 needs to be estimated, e.g. via the sample variance of $\widehat{\varepsilon}$.

Bayesian linear regression

Conjugate priors

- Consider the same linear model:

$$\mathbf{y}|\beta, \sigma^2 \sim N(\mathbf{X}\beta, \sigma^2 I_{n\times n}).$$

- Conjugate priors are

$$\begin{aligned}\beta|\sigma^2 &\sim N(\mathbf{b}_0, \sigma^2\mathbf{B}_0),\\ \sigma^2 &\sim \Gamma^{-1}(s_0, S_0).\end{aligned}$$

- The joint prior is a normal-inverse-gamma (NIG) distribution,

$$(\beta, \sigma^2) \sim \text{NIG}(\mathbf{b}_0, \mathbf{B}_0, s_0, S_0),$$

with density

$$p(\beta, \sigma^2) \propto \frac{\exp(-(\beta-\mathbf{b}_0)^T\mathbf{B}_0^{-1}(\beta-\mathbf{b}_0)/(2\sigma^2) - S_0/\sigma^2)}{(\sigma^2)^{(k+1)/2+s_0+1}}.$$

Linear regression posterior

- The posterior is

$$p(\beta, \sigma^2|\mathbf{y}) \propto \frac{1}{(\sigma^2)^{n/2}} \exp\left(\frac{-1}{2\sigma^2}(\mathbf{y}-\mathbf{X}\beta)^T(\mathbf{y}-\mathbf{X}\beta)\right)$$
$$\times \frac{1}{(\sigma^2)^{(k+1)/2}} \exp\left(\frac{-1}{2\sigma^2}(\beta-\mathbf{b}_0)^T\mathbf{B}_0^{-1}(\beta-\mathbf{b}_0)\right)$$
$$\times \frac{1}{(\sigma^2)^{s_0+1}} \exp\left(\frac{-S_0}{\sigma^2}\right).$$

- Expanding the quadratic form and grouping terms yields

$$(\mathbf{y}-\mathbf{X}\beta)^T(\mathbf{y}-\mathbf{X}\beta) + (\beta-\mathbf{b}_0)^T\mathbf{B}_0^{-1}(\beta-\mathbf{b}_0)$$
$$= \mathbf{y}^T\mathbf{y} + (\beta-\mathbf{b}_n)^T\mathbf{B}_n^{-1}(\beta-\mathbf{b}_n) - \mathbf{b}_n^T\mathbf{B}_n^{-1}\mathbf{b}_n + \mathbf{b}_0^T\mathbf{B}_0^{-1}\mathbf{b}_0,$$

with $\mathbf{B}_n := (\mathbf{X}^T\mathbf{X} + \mathbf{B}_0^{-1})^{-1}$ and $\mathbf{b}_n := \mathbf{B}_n(\mathbf{B}_0^{-1}\mathbf{b}_0 + \mathbf{X}^T\mathbf{y})$.

Linear regression posterior

- Let $s_n := s_0 + n/2$ and

$$S_n = S_0 + \frac{1}{2}(\mathbf{y}^T\mathbf{y} + \mathbf{b}_0^T\mathbf{B}_0^{-1}\mathbf{b}_0 - \mathbf{b}_n^T\mathbf{B}_n^{-1}\mathbf{b}_n).$$

- Collecting everything into the posterior yields

$$p(\beta, \sigma^2|\mathbf{y}) \propto \frac{\exp(-(\beta - \mathbf{b}_n)^T\mathbf{B}_n^{-1}(\beta - \mathbf{b}_n)/(2\sigma^2) - S_n/\sigma^2)}{(\sigma^2)^{(k+1)/2+s_n+1}}.$$

- In other words,

$$\beta, \sigma^2|\mathbf{y} \sim \text{NIG}(\mathbf{b}_n, \mathbf{B}_n, s_n, S_n).$$

- Model fit via e.g. posterior predictive checks: hold out a subset of data and check whether its empirical distribution resembles the posterior predictive density.

Linear regression posterior mean

- We can read off posterior mean regression coefficients from the quadratic form:

$$\mathbb{E}[\beta|\mathbf{y}] = \mathbf{b}_n = \mathbf{B}_n(\mathbf{B}_0^{-1}\mathbf{b}_0 + \mathbf{X}^T\mathbf{y})$$

with

$$\mathbf{B}_n = (\mathbf{X}^T\mathbf{X} + \mathbf{B}_0^{-1})^{-1}.$$

- By using $(A+B)^{-1}A = I_{n\times n} - (A+B)^{-1}B$, we can write

$$\mathbb{E}[\beta|\mathbf{y}] = (I_{n\times n} - \mathbf{A})\mathbf{b}_0 + \mathbf{A}\widehat{\beta}_{\mathsf{LS}},$$

where

$$\mathbf{A} := \mathbf{B}_n\mathbf{X}^T\mathbf{X}.$$

Worked example: Munich rent data

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- Install the `gamlss` package and load the Munich rent data set with `gamlss.data::rent99`.
- Look up the `lm` and `summary` functions and carry out a brief least squares analysis of rent regressed on area, location, as well as the presence of a kitchen and central heating.
- Compute the Bayesian posterior for the same data with prior mean vector $\mathbf{b}_0 = 0$ and prior covariance $\mathbf{B}_0 = 10^5 \cdot I_{n \times n}$.
- Try out other prior values, including extreme ones such as $\mathbf{b}_0 = (10^4, \ldots, 10^4)$ and $\mathbf{B}_0 = 10^{-4} \cdot I_{n \times n}$. Does the posterior mean behave as you expect?

A frequentist linear model

```
library(gamlss)
head(gamlss.data::rent99)
```

	rent	rentsqm	area	yearc	location	bath	kitchen	cheating	district
1	109.9487	4.228797	26	1918	2	0	0	0	916
2	243.2820	8.688646	28	1918	2	0	0	1	813
3	261.6410	8.721369	30	1918	1	0	0	1	611
4	106.4103	3.547009	30	1918	2	0	0	0	2025
5	133.3846	4.446154	30	1918	2	0	0	1	561
6	339.0256	11.300851	30	1918	2	0	0	1	541

```
lm.munich <- lm(rent ~ area + location + bath +
    kitchen + cheating, gamlss.data::rent99)
summary(lm.munich)
```

Frequentist regression output

```
Call:
lm(formula = rent ~ area + location + bath + kitchen +
    cheating, data = gamlss.data::rent99)

Residuals:
Min      1Q      Median  3Q    Max
−633.41 −89.17 −6.26  82.96  1000.76

Coefficients:
             Estimate  Std. Error  t value  Pr(>|t|)
(Intercept) −21.9733     11.6549   −1.885    0.0595 .
area          4.5788      0.1143   40.055   < 2e−16 ***
location2    39.2602      5.4471    7.208  7.14e−13 ***
location3   126.0575     16.8747    7.470  1.04e−13 ***
bath1        74.0538     11.2087    6.607  4.61e−11 ***
kitchen1    120.4349     13.0192    9.251   < 2e−16 ***
cheating1   161.4138      8.6632   18.632   < 2e−16 ***
---
Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' '
    1

Residual standard error: 145.2 on 3075 degrees of freedom
Multiple R−squared: 0.4504,Adjusted R−squared: 0.4494
F−statistic: 420 on 6 and 3075 DF, p−value: < 2.2e−1
```

A Bayesian linear model

```
d = length(coef(lm.munich))
b0 = matrix(rep(0, d))
B0 = diag(10^5, d) # Very high prior variance
X = model.matrix(lm.munich)
y = gamlss.data::rent99$rent
Bn = solve(crossprod(X) + solve(B0))
bn = Bn %*% (t(X) %*% y + solve(B0) %*% b0)
bn

                    [,1]
(Intercept) -21.973323
area          4.578809
location2    39.260218
location3   126.057465
bath1        74.053755
kitchen1    120.434906
cheating1   161.413782
```

Sensitivity to prior parameters

```
d = length(coef(lm.munich))
b0 = matrix(rep(1000, d)) # Misspecified mean
B0 = diag(0.0001, d) # Very low prior variance
X = model.matrix(lm.munich)
y = gamlss.data::rent99$rent
Bn = solve(crossprod(X) + solve(B0))
bn = Bn %*% (t(X) %*% y + solve(B0) %*% b0)
bn

                    [,1]
(Intercept) 9206.9265
area        -283.9473
location2   9165.0774
location3   9954.1464
bath1       9857.9246
kitchen1    9848.3346
cheating1   9016.0030
```